What repeated rhinovirus detections can hide

Main finding: Samples with the same broad virus labels can still contain different genetic sequences. In this dataset, looking beyond species labels revealed differences in 14 of 15 otherwise matching pairs. Repeated detections remain useful; finer genetic detail helps explain what is being detected repeatedly.

Research Process

AI assistance and collaboration. This project used Codex, including parallel AI agents, for literature review, data inspection, coding, analyses, figures, methodological checks and drafting. The participant directed the research questions and requested the comparisons. Human collaborators, individual/team submission status and the participant's personal work times have not yet been confirmed.

The follow-up exploration added whole-cohort sequence and competing-reference checks, exact sample-subset optimization, 400 equal-read optimization analyses, and cross-file checks of a low-identity sequence signal. Their personal activity times remain to be supplied.

Personal time log: pending participant input. The table below reconstructs when saved analysis outputs were produced on 11 October 2026, in Eastern Time. Its windows are 30 minutes long. File modification times do **not** establish how long the participant or an AI agent worked, which papers the participant read, or continuous activity throughout each window. Earlier question development and intervals without documented outputs are not reconstructed.

Output window (ET)	AI-assisted work documented by saved files
20:30–21:00	Four feasibility pilots; VP1 reference analysis; collection-effort and MBTA metadata; sampling-objective analysis; equal-read sensitivity runs. Saved outputs span 20:32–20:59.
21:00–21:30	Equal-read analysis notes, consistency checks and results synthesis. Saved outputs at 21:00, 21:03 and 21:04.
21:30–22:00	Methodological audit, four report figures and the targeted alternative-reference check. Saved outputs at 21:41–21:43.
22:30–23:00	Initial manuscript revision, document exports and layout verification. Saved outputs at 22:35–22:38.

To complete the required process record: add the participant's actual activities and start/end times in blocks no longer than 30 minutes, including collaboration. This retrospective output record is not a substitute for that personal account.

Team background and contributions

Research Writeup

What I wanted to find out

I wanted to understand what repeated rhinovirus detections tell us. When the virus keeps appearing in samples, are we seeing a similar genetic signal again, or different signals grouped under the same broad label?

I analyzed public sequence data from pooled swabs collected around Boston. **Samples with the same rhinovirus species labels often contained different genetic sequences.** Summarizing my results by species hid those differences. This does not show that the surveillance program itself ignores them.

Why these differences could matter

Genetic detail can change our understanding of what is circulating, even when the broad virus label stays the same. Two published studies show why this is useful.

In a Kenyan school study, researchers used genetic typing and sampling dates to classify 109 repeat rhinovirus detections: **85 as new infections and 24 as continuing infections**. A repeated positive result alone could not make that distinction. The study followed individuals; our pooled samples cannot identify individual reinfections. Luka et al., 2020

The geographic evidence is especially relevant here. A later study found that rhinovirus type **C22 appeared and disappeared at similar times across several locations**, while **C38 was detected only in the northern study locations**. Both would be called Rhinovirus C in a species-only summary, hiding their different geographic patterns. Luka et al., 2023

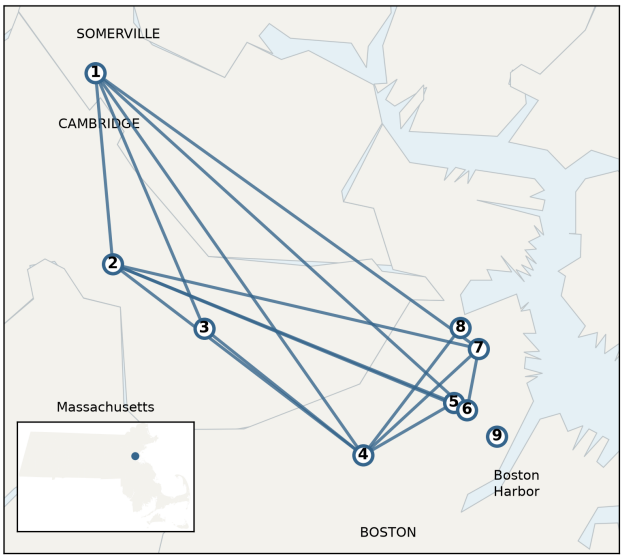
These studies explain why retaining genetic detail can be useful. Our contribution is to measure how much detail a species summary hides in these Boston files. Our reference groups are provisional: we have not shown that their differences change illness severity or identify separate transmission chains. The practical implication is to **keep finer genetic information alongside detection counts when comparing places**, then test whether it improves surveillance decisions.

How the connections between locations change

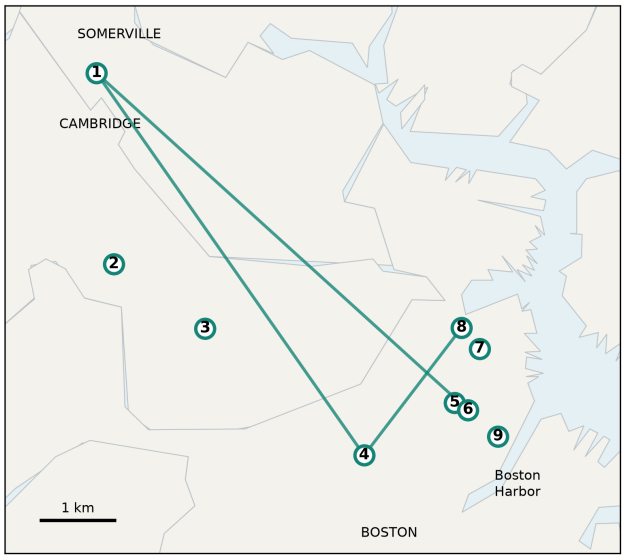
The maps show the same nine Boston-area sampling locations in Massachusetts. A line connects two places when at least one pair of samples collected there during the same week shared a detected label. At the broad species level there are **15 connections**. Requiring a shared finer reference group leaves **4**.

Figure 1 | The same sampling locations, connected at two levels of detail

Broad species labels
15 connected location pairs



Finer reference groups
4 connected location pairs



1 Davis Square
2 Harvard
3 Central Square
4 Copley Square
5 Park / Tremont
6 Downtown Crossing
7 Haymarket
8 North Station
9 South Station

A line means at least one pair of samples collected in the same week shared a detected label.
Both panels use the same 26 eligible sample pairs. Lines do not indicate transmission or travel routes.

Nearby-station positions are approximate location proxies. Seaport had no eligible comparison and is not plotted.
Background: MassGIS municipalities. Coordinates: MBTA. Data: selected June–September 2026 swab files.

Figure 1. Both maps use the same 26 sample pairs with enough sequence evidence in both files. A connection means at least one label was shared in a same-week comparison; the files need not have identical sets of labels. Lines are not transmission paths or travel routes. No line does not establish biological absence. Nearby MBTA station positions approximate the collection areas; Seaport has no eligible comparison and is omitted. The study covers the Boston area, not all Massachusetts.

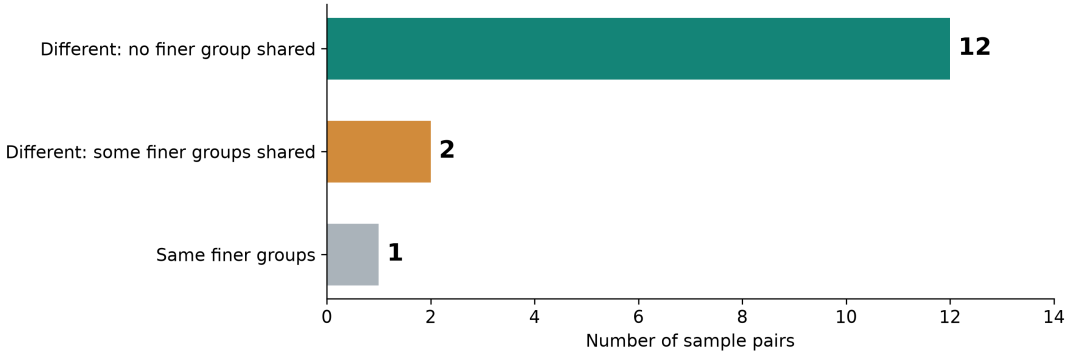
How often did matching broad labels hide a difference?

I compared samples from different locations collected during the same week. In 15 pairs, both samples had exactly the same species labels. Looking more closely at their sequences distinguished **14 of those 15 pairs**. In 12 pairs, none of the finer genetic groups detected in one sample was detected in the other; two pairs shared some groups but differed in others. Only one pair had the same finer groups.

I identified finer differences by matching reads against published viral sequences. I call the resulting categories **reference groups**, not confirmed virus types. A “supported” group has enough matching sequences to pass the rules described in the methods.

Figure 2 | 14 of 15 broadly matching pairs differed at the finer level

Each pair had the same rhinovirus species labels and came from different locations in the same week.



Counts describe sample comparisons, not numbers of viruses or infected people. The 15 pairs reuse 19 files.
Finer labels are provisional reference groups; an undetected group is not proof of biological absence.

Figure 2. Each bar counts sample pairs, not individual viruses or infected people. All 15 pairs had identical broad species labels; 14 differed at finer resolution. These comparisons reuse 19 files and are not independent observations. The map uses any shared label, whereas this chart examines pairs whose complete broad-label lists matched.

To check whether unequal amounts of sequence data explained the result, I kept 500 reads per eligible file and repeated the comparison 200 times. Typically, every pair with matching species labels still differed at the finer level. The middle 95% of repeats ranged from 83.3% to 100%, with 3–9 pairs qualifying each time. This tests the effect of retaining different reads, not uncertainty about the wider community.

Why this matters: repeat detections can strengthen evidence that a virus is circulating. Genetic detail helps us ask whether different places share similar signals or contain distinguishable ones. Published typing studies show that such distinctions can reveal different circulation patterns. Our maps show how the description changes with label resolution; they do not establish who infected whom, how many people were infected, or whether a new outbreak began.

An illustration: what gets lost when we keep only a species checklist?

Of the 48 files analyzed, 31 had enough matching sequence evidence to meet the stricter rules. I used those files for a simple exercise: **how few files would I need to keep if I only wanted a list of which rhinovirus species appeared each week?**

Keeping **14 files** was enough to reproduce that weekly species list. But keeping track of the finer genetic groups required **28 files**. In other words, many files repeated a species name while adding a genetic distinction that the species name did not capture.

The calculation searched for the smallest possible selection of files. I checked the answer separately by trying every possible selection within each week. These are ways of summarizing data already collected, not instructions for how many samples to collect next time.

Even choosing the most genetically informative 14-file selection that preserved the species list retained only **19 of the 28 finer groups** found across the study. At least nine groups—**32%**—would disappear from that shortened summary.

This difference remained under both matching rules and all 200 equal-read repeats for each rule. With stricter matching and equal reads, a typical repeat could omit 41% of files while keeping the weekly species list, but only 4.5% while keeping the finer groups. The corresponding figures with looser matching were 37.5% and 0%.

Repeated samples are not wasted samples. This exercise preserves only whether a label appears in a week. It does not preserve how often it appears, where it appears within that week, or the confidence gained from repeated observations. Those are valuable surveillance outcomes too. The result shows what a short species checklist leaves out; it does not justify cutting collection at a particular location.

Methodology: what data we used and how we compared samples

We analyzed **48 Boston swab-pool sequence files from June–September 2026**, covering ten locations, sampled on a total of 38 location-and-date combinations. A read is one recorded genetic sequence. I used the respiratory viral reads released through SecureBio's Zephyr dashboard. They are a selected portion of the sequencing output, not complete raw sequencing data or individual patient records.

The earlier exploratory pilot selected 12 files per month using a fixed random seed (20261011), excluding three separate pilot files. The full files contain 1,357,509 released reads. To keep the analysis manageable, I randomly retained up to 2,000 reads per file with reproducible file-specific seeds, giving **50,865 reads** to analyze. Short files contributed all available reads. File selection was not balanced across locations, collection dates or pool sizes.

Sampling overlapped across locations within 11 calendar weeks, but never on the same exact date in this 48-file dataset. That limits what we can conclude about differences between locations.

The analysis had three steps:

1. **Compare the same part of the virus.** I compared reads with published sequences for VP1, a gene region encoding part of the virus’s outer shell that is used to distinguish rhinoviruses. The panel contained 1,424 reference intervals in 143 normalized label groups. Only 59 intervals had explicit VP1 annotations; 1,365 were inferred by protein-sequence alignment. Formal classification requires additional genetic interpretation. McIntyre et al., 2013

2. **Count a group only when enough sequences match.** The main, stricter rule required at least 95% alignment identity, a 750-nucleotide alignment, 85% coverage of the reference VP1 interval and five distinct sequence strings per file/group. Duplicate strings were counted once. These thresholds are exploratory evidence rules, not validated diagnostic or genotype criteria.

3. **Compare broad labels with finer genetic detail.** I first recorded the reference groups detected in each file, then summarized those same results as rhinovirus species A, B or C. I checked whether files with the same species labels also had the same finer groups, and repeated the checks using equal numbers of reads and looser matching rules.

A check using two samples from the same place and day

Two Downtown Crossing files dated August 14, each listed as a 50-swab pool, contained 942 distinct sequences matching an A49-labelled group and 795 matching an A1-labelled group. Both groups belong to Rhinovirus A. Removing the original dominant reference sequences preserved 942 and 783 supporting sequences. All **1,737** original supporting sequences also retained a 10% matching advantage when checked against references outside their assigned group.

The labels remain provisional, and a sparse panel with one reference per group recovered neither supported call. Database coverage therefore matters. The detailed methods audit and reference check retain the full tests. These differences do not establish differences in severity or transmission.

Did this tell us where to sample?

The results did not identify a reliably better sampling location. Some places looked more informative when I counted finer groups instead of species. But their order changed when I changed the matching rules or gave each file the same number of reads.

For example, among pools listed as containing 90 swabs, Copley averaged 1.17 species and 1.17 reference groups per file; Park/Tremont averaged 1.00 species and 1.25 groups. Their order reverses, but the files differ in dates and available reads. This alone is insufficient to recommend a location.

When we required both 90-swab pools and equal retained read counts, only three sites remained eligible. At least one strict reversal occurred in **1.5%** of repeats with stringent filters, versus **80.5%** with the less stringent filters. That sensitivity is the reason we do not claim that finer analysis identifies a different best site.

We also joined locations to nearby MBTA station-entry data. Among eight sites with at least two eligible June–August files, the association between entries and average label richness was the same at both resolutions (Spearman correlation −0.395). Requiring three files left six sites and weakened it to −0.086. Too few locations were available, and their sampling conditions differed, to conclude that busy transport hubs were better places to sample. Station entries are an imperfect proxy for the people actually sampled, and we did not analyze residential population density. MBTA data

What “best” could mean

We tested several sampling objectives because they answer different questions. Operational results use collection metadata; sequence-diversity results use supported assignments from the selected files.

Objective	What we could measure	What it tells us
Most positive samples per hour	Pools with any reported viral reads: Haymarket 0.696/hour; Park/Tremont 0.547/hour across reconciled June–September dashboard days	Pool size changes this metric. It does not measure infected individuals per hour.
Most distinct groups per fixed swab budget	Compared declared 60- and 90-swab pools separately, then checked equal-read sensitivity	We can compare intact observed pools; we cannot computationally divide the people in a pool. No robust best site emerged.
Most additional groups beyond other sites	Harvard added an average of 0 species but 0.80 reference groups per observed multisite week	Finer labels can reveal added information, but dates and within-day effort differ.
Earliest detection	First observed file for nine supported groups appearing at two or more sites	First in this dataset is not the true introduction date or demonstrated early warning.
Broadest two-site coverage	Park/Tremont plus Copley had the highest unstandardized expected group count, 2.43, among eligible pairs	This is a retrospective calculation with unequal dates, pool sizes and read counts, not a validated network recommendation.

Hours were inferred from reported swabs/hour; they are not staff-hours or laboratory-hours. The full dashboard audit recovered 502 pool rows, including 16 zero-read rows missing from a download-link-only manifest. Pool-based analyses requiring complete sequence coverage and reconciled collection totals had only 14 qualifying location-days and 21 files. The broader collection-rate comparisons above use the reconciled dashboard records, not just those 21 files.

For weekly additional information, we enumerated one observed day per site within a week and counted labels present at that site but absent elsewhere. This balances observed day opportunities, not swab or read effort. For two-site coverage, we averaged the number of different labels obtained by choosing one existing file at each site. Neither calculation was tested prospectively.

What this study adds to existing work

Rhinovirus diversity across social and geographic settings is already studied, and network-based surveillance design is an established topic. Luka et al., 2023 Pei et al., 2021 This study asks what repeated detections mean in these public pooled-swab files: when the same species name appears again, does it describe similar sequence information or conceal a difference? It also checks whether finer information changes comparisons between sampling locations.

Looking more closely naturally creates more possible categories. The useful result is how often apparently matching samples differed, whether those differences survived further checks, and whether they supported a practical conclusion about where to sample. The evidence for genetic differences was stronger than the evidence for changing locations.

Technical details and limitations

Alignment and reference construction. We used mappy 2.31 (map-ont, k=11, w=5, best_n=500) and selected the returned alignment with most matching bases. Identity was matching bases divided by alignment block length. Reference-span coverage was the covered interval divided by VP1 reference length. VP1 annotation transfer used BLOSUM62 protein alignment, gap penalties 10/0.5 and ≥95% peptide-span coverage, preferring same-label seeds before same-species seeds. A1/A1A/A1B labels were merged as A1_complex. Reference annotations came from GenBank.

Filters and sensitivity. The less stringent filter required 90% identity, 600 aligned nucleotides and 70% reference span. Both filters required five distinct strings per file/group. A heuristic alternative-label matching margin was also applied (5%/10%), but 11,192 of 11,194 stringent retained read rows had no other-label alternative returned and therefore a margin of 1.0. It was effectively non-discriminating. A subsequent whole-cohort audit removed each winning reference group before remapping: 11,171 of 11,194 distinct read rows retained the 10% matching margin. All 47 supported file/group associations and all 28 supported groups remained. Returned alignments are still heuristic, not exhaustive specificity validation. Stringent filtering supported 28 groups in 31 files; base filtering supported 36 groups in 33 files.

Equal-read analysis. We sampled 500 retained reads per eligible file, without replacement, for 200 repeats (seed 20261012), including unassigned reads. We then removed duplicate strings and reapplied the five-string rule. Twenty-nine files qualified; site rankings required at least two eligible files per site. A strict reversal required opposite nonzero pairwise score differences. No confirmatory significance tests were used.

A recurring signal that our original matching rule missed. I built a summary sequence from many reads in one July 17 file. That sequence was only about 90% identical to its closest published sequence in our comparison database, below the original 95% cutoff. But it closely matched reads in four other files: 282 of 284 passed the same identity, length and coverage requirements when compared with the summary sequence. Those other files were not used to build it. This suggests that a similar signal recurred and that our database lacked a close enough match. It does not establish a new virus type or a transmission link.

Limits on interpretation. The 15 headline comparisons involve only 19 unique files, so they are not independent. An undetected group may reflect limited reads or reference coverage rather than biological absence. Same-day files have not been verified as independent biological pools. No different locations were sampled on the same exact date in this cohort. We did not analyze diagnostic controls, reconstruct whole sample genomes, validate genotypes phylogenetically, estimate infection prevalence or measure early-warning lead time. The research question evolved after preliminary data inspection; there was no independent confirmatory set.

Conclusion. Repeated rhinovirus detections can contain both recurring genetic signals and differences hidden by a shared species name. In this dataset, finer comparisons distinguished 14 of 15 pairs that looked the same at species level. A separate check also recovered a recurring signal that the original matching rule missed. Published studies establish that genetic typing can help distinguish new infections and geographic circulation patterns; our analysis shows how much potentially useful detail a broad summary can hide. The next step is to validate these reference groups and compare locations sampled on matched dates with comparable effort. These findings support retaining genetic detail alongside repeat-detection counts, but do not establish prevalence, transmission routes or a reason to stop sampling at any particular location.

Code, data and references

The maps use MassGIS municipal boundaries and coordinates from the MBTA stops API. Raw geographic data, station-to-site proxy choices and download URLs are saved in data/map/ and report/figure_data/map_nodes.csv. Map edges and all underlying sample comparisons are exported alongside the figures.

The new exploration and checks, local codebase, analysis protocol, methods audit and figure source data accompany this report. The project reuses inputs from the neighboring zephyr-depth-audit directory and is not yet a standalone public repository. **Public repository URL: pending.** No external submission has been made.

The new budget analyses are in src/discovery_sample_budget.py and src/discovery_budget_sensitivity.py; independent checks are in src/discovery_validate.py. The two report figures are generated by src/report_two_figures.py. Core scripts are src/typing_vp1.py, src/typing_sensitivity.py, src/sampling_objectives.py, src/objectives_equal_reads.py and src/report_competitor_check.py. Dependency versions, figure data and SHA256 hashes are included in the report folder. Computational checks verify accounting and reproducibility, not biological ground truth.

1. McIntyre CL, Knowles NJ, Simmonds P. Proposals for the classification of human rhinovirus species A, B and C into genotypically assigned types. *J Gen Virol*. 2013;94:1791–1806. doi:10.1099/vir.0.053686-0.

2. Luka MM, et al. Rhinovirus dynamics across different social structures. *npj Viruses*. 2023;1:6. doi:10.1038/s44298-023-00008-y.

3. Pei S, Teng X, Lewis P, Shaman J. Optimizing respiratory virus surveillance networks using uncertainty propagation. *Nat Commun*. 2021;12:222. doi:10.1038/s41467-020-20399-3.

4. SecureBio. Zephyr dashboard. Snapshot accessed 11 October 2026.

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